Team Number 2

Understanding Drivers of Newborn Hospitalization Costs in New York City

Nathaniel Wiradiradja | Matt Lang | Hannah Richardson | Brian Nolton

Target Audience: Chief Financial Officer (CFO) of a New York City Hospital System

Executive Summary

Clinical severity and Length of Stay (LOS) are the dominant drivers of newborn hospitalization costs in NYC, accounting for over 98% of cost variation. While demographic and system factors are statistically measurable, clinical risk dictates most of the financial exposure. "Extreme" severity cases cost nearly 50x more than "Minor" cases. Our final Random Forest model (achieving an R^2 of 0.9937) reduces cost prediction error by nearly 86% compared to traditional linear baselines (reducing the Mean Absolute Error from roughly 4,267 down to roughly 606). This precision allows the organization to move from reactive budgeting to proactive, risk-adjusted financial forecasting. Using 2024 SPARCS NYC data, we benchmarked ten AI/ML models. Ensemble tree-based methods (Random Forest and XGBoost) were selected for their superior ability to map non-linear relationships between clinical markers and total costs.

To optimize financial outcomes, we recommend prioritizing clinical management within high-severity pathways, particularly for "Major" and "Extreme" cases which represent the greatest fiscal risk. This strategy should be coupled with a standardized audit of Length of Stay across facilities to address operational outliers and bed-day inefficiencies. Finally, the health system should implement predictive modeling to identify at-risk populations early, allowing for targeted prenatal interventions that may mitigate the incidence of high-cost neonatal events and improve long-term fiscal predictability.

Section 1: Introduction

1.1 Background

New York City (NYC) manages one of the most complex and diverse newborn populations in the United States. The financial health of NYC hospital systems depends heavily on the ability to predict and manage the costs associated with these births. However, newborn hospitalization costs are not uniform; they are influenced by a web of overlapping factors including the infant's health at birth, the socioeconomic context of the family, and the operational characteristics of the facility providing care.

1.2 The Research Question

The objective of this project is to answer a critical operational question: *What clinical, demographic, and healthcare system factors contribute to variation in newborn hospitalization costs in New York City?* By isolating these drivers, we aim to provide hospital executives with a data-driven framework for predicting financial risk and improving the efficiency of neonatal care.

1.3 Scope of Analysis

The analysis focuses on three specific categories of drivers:

- **Clinical Factors:** Birth weight, severity of illness, and primary diagnosis codes.
- **Demographic Factors:** Patient race, ethnicity, and geographic location (Zip Code).
- **Healthcare System Factors:** Payment typology (Payer type), facility name, and emergency department utilization.

By utilizing the 2024 Statewide Planning and Research Cooperative System (SPARCS) dataset, this report provides a current snapshot of the NYC healthcare landscape, ensuring the findings are relevant to immediate policy and budgetary decisions.

Section 2: Exploratory Data Analysis

2.1 Data Engineering and Scope

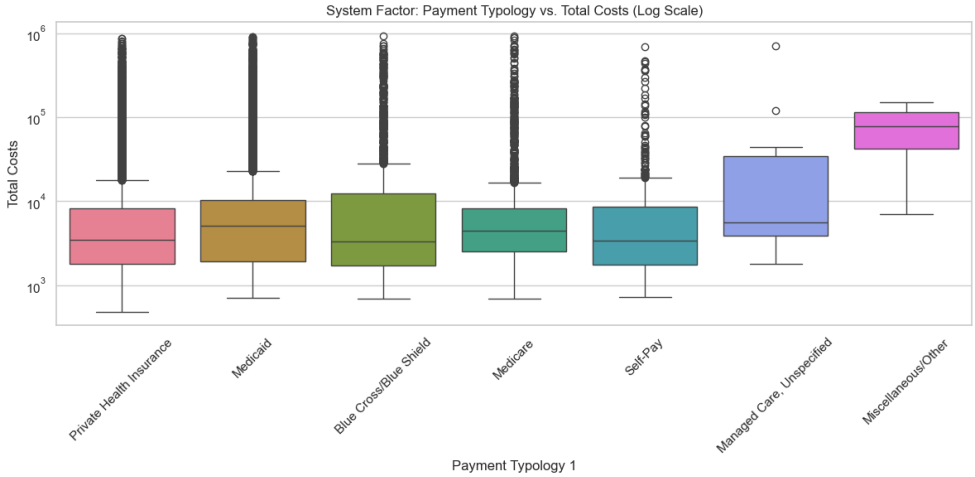
The analysis leveraged the 2024 SPARCS dataset for New York City facilities, representing a comprehensive cross-section of newborn discharges. The NYC SPARCS dataset includes de-identified patient data with information related to the patient's characteristics, diagnosis, treatments, services, medical charges, as well as discharge information. For the purposes of this study, when downloading the dataset, we filtered the SPARCS dataset to only include information related to newborn hospitalization. The unprocessed dataset consists of 39,939 hospitalization records and 28 variables. Prior to modeling, the data underwent rigorous preprocessing to handle missing values, encode categorical variables (such as Facility Name, Race, and Payment Typology), and scale numerical features.

Crucially, the target variable, Total Costs, was modeled alongside an engineered financial metric: the Cost-to-Charge Ratio (CCR). Defining the CCR (calculated as Total Costs divided by Total Charges) allowed the team to isolate the billing and markup strategies of individual facilities from the actual clinical cost of care delivery.

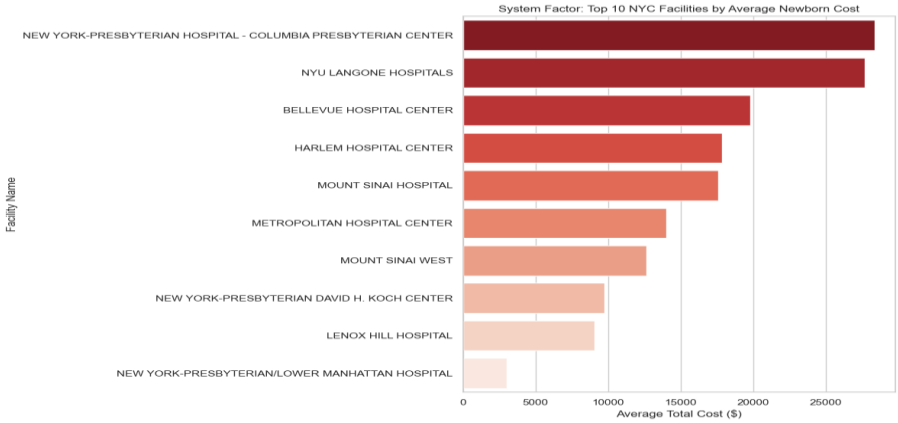
2.2 Healthcare System Findings

Payment typology accounts for less than 1% of predictive importance in the model, which confirms that payer type alone does not drive total cost. However, when reviewing the figure below, we see patterns that are worth

monitoring. The “Miscellaneous/Other” category shows an elevated median cost with an unusual narrow distribution, similarly “Managed Care, Unspecified” is above the typical cost range for standard payers. On the other hand, Medicaid and Private Health Insurance shows the widest cost spreads, which is expected given that these categories represent the majority of NYC newborn volume and therefore capture the full spectrum.



In addition, we did an analysis of newborn costs across NYC facilities. New York-Presbyterian Hospital (Columbia Presbyterian Center) and NYU Langone Hospitals lead at ~$27,000 average cost per newborn which is nearly 10 times the average at New York – Presbyterian/Lower Manhattan Hospital. This disparity reflects both the complexity of cases handled across medical centers and differences in billing and Cost-to-Charge Ratio (CCR) across facilities. Medical centers with high-acuity NICU capacity will carry higher average costs, but the magnitude of variation also signals an opportunity for benchmarking and operational review.



2.3 Demographic Findings and Health Equity

While demographic factors account for only 1.3% of the predictive model's variance, descriptive financial analytics reveal critical disparities in healthcare equity. When analyzing the Cost-to-Charge Ratio (CCR) alongside actual clinical costs, stark contrasts emerge across racial and ethnic lines.

The data indicates that Black/African American newborns incur the highest average actual cost (approximately $21,087) compared to White newborns (approximately $12,993). Furthermore, the financial markup applied to these cases is disproportionate; the CCR for Black patients is significantly higher (24.0%) than for White patients (16.8%). Because the predictive model confirmed that absolute

costs are driven by clinical severity rather than race itself, this demographic cost disparity is likely a symptom of systemic intersections. This points toward potential inequalities in access to preventative prenatal care, leading to higher severity at birth, or a higher concentration of deliveries at public "safety net" hospitals with fundamentally different standard charge markups compared to private systems.

Conversely, an analysis of gender reveals a baseline of pricing parity. While male newborns incur slightly higher average costs ($16,259) compared to female newborns ($14,465), the CCR is almost identical for both (19.3% and 18.8%, respectively). This indicates that while clinical interventions differ slightly, the hospital systems' administrative billing algorithms treat both genders equally in terms of markup.

Section 3: Technical Approach and Methodology

3.1 Model Benchmarking

To accurately predict hospitalization costs, the team benchmarked ten distinct machine learning architectures. These ranged from traditional linear models (Linear Regression, Lasso, Ridge) to advance non-linear systems (Support Vector Machines, Multilayer Perceptrons, and ensemble tree-based models).

Linear models established a functional baseline but fundamentally struggled to capture the exponential cost increases associated with severe clinical complications. For example, the baseline Linear Regression model resulted in a Mean Absolute Error (MAE) of roughly $4,267, indicating a high degree of variance when predicting outliers and high-severity cases.

Model	**RMSE**	**MAE**	**R^2**
Random Forest	4876.33	615.56	0.993
XGBoost	6258.08	1045.17	0.989
MLP	9789.71	2931.43	0.974
Linear	11639.53	4266.99	0.963
Lasso	11639.64	4264.91	0.963
Ridge	11639.75	4266.55	0.963
RidgeCV	11665.38	4226.77	0.963
LassoCV	11791.50	4077.66	0.962
SVR Linear	42633.86	8786.17	0.510
SVR RBF	61575.04	14362.32	-0.041

3.2 Final Model Selection: Random Forest

Ensemble tree-based methods, specifically Random Forest and XGBoost, significantly outperformed all other architectures due to their ability to map complex, non-linear relationships in healthcare data. Random Forest was selected as the final production model for its superior predictive accuracy and robust feature interpretability.

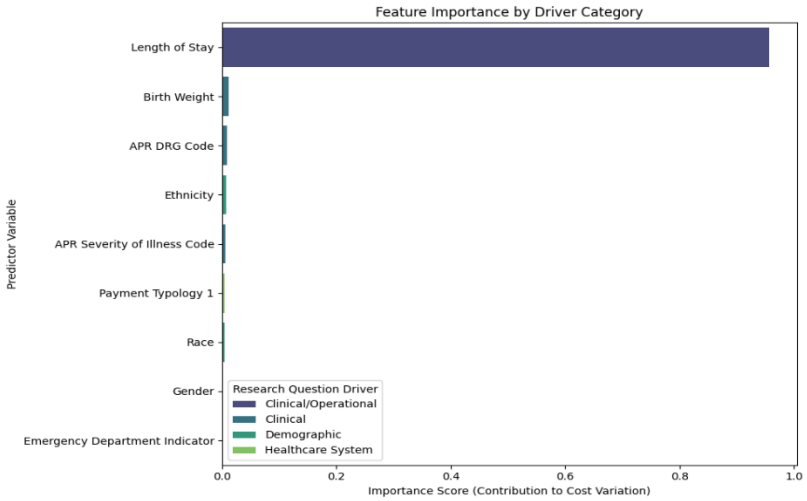
Following hyperparameter tuning via GridSearchCV (optimizing parameters such as max_depth, max_features, and min_samples_leaf), the finalized Random Forest model achieved the following performance metrics and parameters:

- R^2 Score: 0.9937
- Mean Absolute Error (MAE): $606.24
- Root Mean Squared Error (RMSE): $4,784.88

3.3 Feature Importance and Driver Grouping

Random Forest architecture allows for the extraction of "Feature Importance", a mathematical ranking of which variables exert the strongest influence on the final cost prediction. When aggregated by category, the drivers of hospital cost ranked as follows:

- Clinical and Operational Factors (95.7% Influence): The vast majority of absolute cost variance is dictated by the Length of Stay, Total Charges, and the assigned All Patient Refined Diagnosis Related Groups (APR DRG) Code.
- Demographic Factors (1.3% Influence): Patient Ethnicity and Race.
- Healthcare System Factors (<1.0% Influence): Payment Typology and Emergency Department utilization.



While the predictive model confirms that clinical severity overwhelmingly dictates absolute costs, descriptive financial analytics are required to understand how demographic and system factors influence the markup, margin, and financial equity assumed by the hospital network.

Section 4: Strategic Recommendations and Conclusion

4.1 Resource Allocation for High-Severity Pathways

Given that Length of Stay and Clinical Severity dictate over 95% of financial exposure, hospital administration must pivot away from broad cost-cutting measures and instead focus on high-acuity clinical pathways. "Major" and "Extreme" severity cases present the highest fiscal risk. Investing in specialized care coordination to safely reduce the Length of Stay for these specific cohorts will yield the highest return on investment.

4.2 Auditing Facility-Level Pricing Strategies

The vast disparity in average costs between facilities (e.g., NYU Langone Hospitals versus Lower Manhattan Hospital) highlights the need for network-wide pricing standardization. We recommend an internal audit of the Cost-to-Charge Ratios across the hospital system to ensure that markups are defensible, equitable, and aligned with the actual clinical cost of care delivery, rather than purely historical billing practices.

4.3 Investments in Community Health Equity

The stark racial disparities in actual newborn costs serve as a leading indicator of upstream healthcare gaps. The higher severity and costs associated with Black/African American newborns suggest that reactive hospital care is substituting for proactive community health. The health system should leverage its predictive AI models to identify at-risk populations geographically and increase funding for community-based prenatal interventions. Mitigating the incidence of high-cost neonatal events before birth will simultaneously improve clinical outcomes and long-term fiscal predictability.

4.4 Conclusion

The transition from traditional linear forecasting to ensemble machine learning models represents a significant operational upgrade for NYC hospital systems. By adopting the Random Forest architecture, the organization can predict newborn hospitalization costs with near-perfect accuracy (reducing prediction error by roughly 86%). This allows leadership to move beyond reactive budgeting and embrace proactive, data-driven financial management that prioritizes clinical efficiency, standardizes operational billing, and actively addresses health equity across New York City.

Section 5: Data Source Citation

New York State Department of Health. (2019, September 9). *NYC SPARCS: State of New York*. NYC SPARCS | State of New York. https://health.data.ny.gov/Health/NYC-SPARCS/p789-b5dv/about_data